

Low Risk of Primary *Clostridium difficile* Infection With Tetracyclines: A Systematic Review and Metaanalysis

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Low Risk of Primary *Clostridium difficile* Infection With Tetracyclines: A Systematic Review and Metaanalysis

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Background. The choice of antibiotics for systemic infections in patients with a high risk of *Clostridium difficile* infection (CDI) remains a clinical practice dilemma. Although some studies suggest that tetracyclines may be associated with a lower risk of CDI than other antibiotics, other results are conflicting. We conducted a systematic review and metaanalysis of studies that assessed the risk of CDI with tetracyclines compared to other antibiotics.

Methods. We conducted a systematic search of Medline, Embase, and Web of Science from January 1978 through December 2016 to include studies that assessed the association between tetracycline use and risk of CDI. Weighted summary estimates were calculated using generalized inverse variance with a random-effects model using RevMan 5.3. Study quality was assessed using the Newcastle-Ottawa scale.

Results. Six studies (4 case control, 2 cohort) with patient recruitment between 1993 and 2012 were included. Metaanalysis using a random-effects model, demonstrated that tetracyclines were associated with a decreased risk of CDI (odds ratio [OR], 0.62; 95% confidence interval [CI], 0.47–0.81; $P < .001$). There was significant heterogeneity, with an I^2 of 53% with no publication bias. Subgroup analysis of studies that evaluated the risk of CDI with doxycycline alone also demonstrated a decreased risk of CDI (OR, 0.55; 95% CI, 0.40–0.75; $P < .001$).

Conclusions. Metaanalyses of existing studies suggest that tetracyclines may be associated with a decreased risk of CDI compared with other antimicrobials. It may be reasonable to use tetracyclines whenever appropriate to decrease CDI associated with antibiotic use.

Keywords. *Clostridium difficile* infection; incidence; metaanalysis; tetracyclines.

Clostridium difficile is the leading cause of healthcare-associated infection in the United States, accounting for almost half a million cases, 29 300 deaths, and \$4.8 billion in excess medical costs annually [1, 2]. Well-established risk factors for *C. difficile* infection (CDI) include older age, hospitalization, and use of antibiotics and possibly proton-pump inhibitors [3, 4].

Antibiotic use is the most common risk factor for CDI. This risk increases with antianaerobic spectrum, multiple concomitant antibiotics, and prolonged duration of use [5]. Antibiotics associated with a high risk of CDI are fluoroquinolones, clindamycin, broad-spectrum cephalosporin, and penicillin. However, patients who receive aminoglycosides, metronidazole, vancomycin, and tetracyclines may be at lower risk for CDI [6, 7]. Antibiotics disrupt the host colonic microbiome and impair the colonization resistance afforded by it. Tetracyclines inhibit protein synthesis by binding with the bacterial 30S ribosomal

subunit [8]. They have excellent bioavailability and tissue penetration [8]. In addition to antimicrobial activity, tetracyclines also have antiinflammatory properties that might contribute to their therapeutic benefits [9]. Tetracyclines are used in various clinical settings for treatment of both gram-positive and gram-negative bacterial infections. Doxycycline is the most commonly used tetracycline, with clinical use similar to the general indications of tetracyclines. Other US Food and Drug Administration–approved uses for doxycycline include malaria prophylaxis, syphilis treatment in penicillin-allergic patients, and for severe acne. Doxycycline is also used off-label for acute bacterial rhinosinusitis, pelvic inflammatory disease, proctitis, periodontitis, vancomycin-resistant enterococcal infections, pneumonia, and flares of chronic obstructive pulmonary disease, and as a sclerosing agent in pleural effusions [10].

Unfortunately, patients who are at high risk for CDI are also at risk for other infections such as recurrent urinary tract infections and upper respiratory infections, because of the presence of underlying comorbid conditions. A common conundrum in clinical practice is to identify and choose antibiotics that may be associated with lower risk or be protective for CDI while still being effective. Several studies have assessed the role of tetracyclines and suggested a lower risk of CDI compared with other antibiotics [11, 12]. One case-control study demonstrated a 27%

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decreased risk of CDI among patients who received doxycycline plus ceftriaxone compared with ceftriaxone alone [11]. Another study found that tetracyclines had no effect on the development of CDI (odds ratio [OR], 0.9; 95% confidence interval [CI], 0.5–1.5) [12]. We hypothesized that tetracyclines may confer a decreased risk of CDI compared with other antibiotics and may be considered as antibiotics of choice (when indicated) for patients at risk for CDI. Given conflicting results, we conducted a systematic review and metaanalysis to determine the association between tetracyclines and the risk of CDI.

METHODS

We followed the MOOSE (Meta-analysis Of Observational Studies in Epidemiology) criteria [13] for observational studies to conduct this metaanalysis.

Selection Criteria

The studies considered for inclusion were case-control studies, cohort studies, and clinical trials. For case-control studies, cases were defined as patients with CDI, and controls were defined as patients without CDI. All cases had prior antibiotic exposure, qualified as tetracycline use, whereas controls were exposed to antibiotics other than tetracyclines or no antibiotics [12, 14–16]. For cohort studies, the study population comprised patients who were evaluated for occurrence of CDI after receiving tetracyclines compared with other antibiotics or no antibiotics, in a defined time period [11, 17]. There was no restriction on study setting (inpatient or outpatient). Studies that did not evaluate CDI as an outcome or did not include cases with prior antibiotic exposure were excluded. If data were lacking to determine an estimate of an OR and 95% CI, these studies were also excluded from metaanalyses. Only published full-text studies were included.

Data Sources and Search Strategy

A comprehensive search of electronic databases from January 1978 through December 2016 was conducted. They included Ovid Embase, Ovid Cochrane Central Register of Controlled Trials, Ovid Medline In-Process & Other Non-Indexed Citations, Ovid Medline, Web of Science, Scopus, and Ovid Cochrane Database of Systematic Reviews. The search strategy was designed and conducted by the Mayo Clinic library staff and 2 study investigators (S.K. and R.T.). Controlled vocabulary, supplemented with keywords, was used to search for studies of tetracycline use and CDI. The main keywords used in the search were *Clostridium difficile*, *C diff*, *C difficile*, *Clostridium difficile* infection, CDI, *Clostridium difficile*-associated diarrhea or CDAD, or pseudomembranous colitis AND Tetracycline OR Chlortetracycline OR Demeclocycline OR Doxycycline OR Lymecycline OR Methacycline OR Minocycline OR Oxytetracycline OR Rolitetracycline AND outcomes, infection. The search was limited to English-language studies. The detailed search strategy is shown in the Supplementary Materials.

The titles and abstracts of the identified studies were reviewed by 2 authors (S.K. and R.T.); studies that did not address the research question were excluded. For the remaining articles, the full texts were reviewed to assess inclusion criteria fulfillment. The reference lists of included articles were also reviewed for additional pertinent studies.

We used the Newcastle-Ottawa scale to assess the methodologic quality of included studies [18]. This scale scores observational studies across 3 categories: patient selection (4 questions), comparability (1 question), and ascertainment of the outcome of interest (3 questions). For patient selection and outcome of interest, 1 point was given if the study met the criterion; for comparability of study groups, 2 points were given if the study controlled for age, sex, or both and other confounding factors (Supplementary Table S1). Studies with a cumulative score of 7 or greater were considered high quality. Discrepancies were addressed by joint reevaluation of the original article.

Data Abstraction

Data were independently abstracted to a defined collection form by 2 authors (S.K. and R.T.). Data collected for each study included study setting and design, year of publication, location, and primary outcome reported. Conflicts in data abstraction were resolved by consensus, referring to the original article.

Outcomes Assessed

Our primary analysis focused on assessing the OR for the risk of CDI after exposure to tetracyclines compared with other antibiotics.

Statistical Analyses

All individual studies reported adjusted OR, which was used in the analysis. Weighted summary estimates were calculated using generalized inverse variance with the random-effects model, as described by DerSimonian and Laird [19]. Heterogeneity within groups was assessed with the I^2 statistic, which estimates the proportion of total variation across studies that is due to heterogeneity in study patients, design, or interventions rather than chance; I^2 values greater than 50% suggest substantial heterogeneity [20]. The presence of publication bias was assessed by visual inspection of funnel plots; however, the results of tests for publication bias on fewer than 10 included studies should be interpreted with caution. All P values were 2-tailed. For all tests (except for heterogeneity), a probability level less than .05 was considered statistically significant. To explain differences between subgroups, a test for interaction was performed ($P_{\text{interaction}}$); a value less than .1 was considered significant for difference between subgroups [21]. Calculations were performed and graphs constructed using RevMan, version 5.3 (the Cochrane Collaboration).

A priori-defined sensitivity analyses included subgroup analyses based on study design and studies that specifically assessed the risk of CDI with doxycycline. To explore the causes

of heterogeneity, we also performed subgroup analyses based on study setting and CDI diagnosis method.

RESULTS

Search Results

The literature search returned 480 potentially relevant studies; titles and abstracts were screened, and full papers were obtained for relevant articles (Figure 1). A total of 38 full-text articles were reviewed, of which 32 were excluded for various reasons. Six full-text studies were included in this metaanalysis: 4 case-control and 2 cohort studies [11, 12, 14–17].

Quality of Included Studies

The median Newcastle-Ottawa quality score for both case-control studies and cohort studies was 7 (range, 7–8) out of 9 points (Supplementary Table S1).

Characteristics of Included Studies

The characteristics of the 6 included studies are shown in Table 1. For the case-control studies, cases were defined as patients with CDI; controls were defined as patients who did not have CDI and who were evaluated for the occurrence of CDI with tetracyclines compared with other antibiotics (no tetracyclines) or no antibiotics [12, 14–16]. For the cohort studies, the study populations included patients who received tetracyclines compared with other antibiotics or no antibiotics in a defined time period and were evaluated for the occurrence of CDI [11, 17]. One study included all patients who received ceftriaxone in the past 30 days and evaluated the occurrence of CDI with ceftriaxone plus doxycycline vs ceftriaxone alone [11].

Four studies were performed in the United States, and 1 each in the United Kingdom and Canada. The earliest study

recruitment period began in 1993 and the latest ended in 2012. All observational studies assessed medication exposure through review of medical records. Index date was the date of diagnosis of CDI, and cases and controls were contemporary in individual studies. The timing of antibiotic exposure before the index date ranged from 30 to 180 days. All cases and controls had similar time periods during which they were assessed for antibiotic exposure before the index date. For multivariate analysis, all studies adjusted for the use of gastric acid suppression medications. All but 1 study also adjusted for age, sex, and recent hospitalization (Supplementary Table S2).

Tetracycline Use and CDI Occurrence

A total of 5322 CDI cases and 412 208 controls were included (available in 5 studies; 1 study did not report total number of patients; Supplementary Table S3). Metaanalysis of all 6 studies using the random-effects model with adjusted ORs demonstrated that tetracyclines were associated with a decreased risk of CDI compared with other antibiotics (OR, 0.62; 95% CI, 0.47–0.81; $P < .001$; Figure 2A). There was significant heterogeneity among the studies, with an I^2 of 53%. No publication bias was seen (Figure 2B).

Doxycycline Use and CDI Occurrence

Three of the 6 studies evaluated the risk of CDI with doxycycline. Subgroup analysis of these studies revealed that doxycycline was associated with a decreased risk of CDI compared with other antibiotics (no doxycycline) (OR, 0.55; 95% CI, 0.40–0.75; $P < .001$; Figure 3).

Subgroup Analyses

Given the significant heterogeneity in the metaanalysis of all 6 included studies, we performed subgroup analyses to better understand the heterogeneity.

Study Design

Subgroup analyses based on study design, separating cohort and case-control studies, revealed that tetracyclines had no significant effect on the risk of CDI in the 4 case-control studies (OR, 0.68; 95% CI, 0.40–1.15; $P = .15$; Figure 4). However, tetracyclines were associated with decreased risk of CDI among the 2 cohort studies (OR, 0.60; 95% CI, 0.41–0.87; $P = .007$; Figure 4).

CDI Definition

Of the 6 studies, 2 used *International Classification of Diseases, Ninth Revision* (ICD-9) code 008.45 for CDI diagnosis, and 4 used a clinical definition (presence of diarrhea with a positive stool test; Table 1). Subgroup analysis of the studies based on these definitions demonstrated a significantly decreased risk of CDI with use of tetracyclines in the studies using the clinical definition (OR, 0.59; 95% CI, 0.44–0.80; $P < .001$) but not among the studies that used ICD-9 codes (OR, 0.95; 95% CI, 0.45–2.01; $P = .90$; Figure 5).

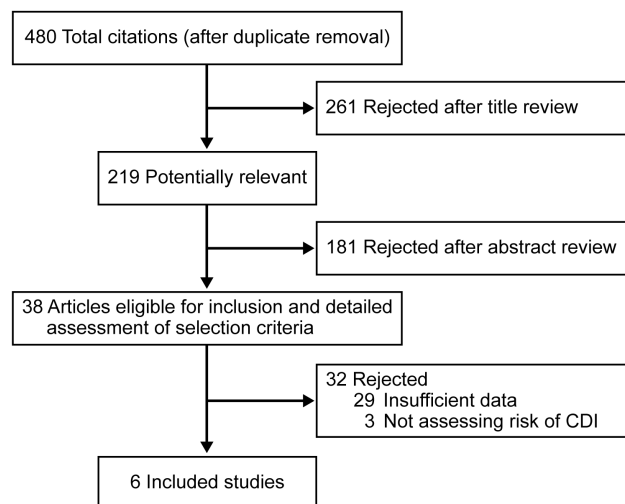


Figure 1. Flow diagram of study selection process. Abbreviation: CDI, *Clostridium difficile* infection.

Table 1. Characteristics of Included Studies

Study, Year	Study Design	Study Setting	Mode of CDI Acquisition	Location	Years	CDI Definition	Tetracyclines Used	Antibiotic Exposure Before Index Date, d	Definition of Controls
Baxter et al, 2008 [14]	Case-control, retro	Hospital (inpatients only)	Hospital-acquired	United States	1999–2005	Diarrhea plus positive CDI toxin (ELISA)	Doxycycline	60	No CDI + antibiotic exposure (other than tetracycline) within 60 d
Delaney et al, 2007 [12]	Case-control, retro	Hospital (inpatients and outpatients)	Community-acquired	United Kingdom	1993–2004	Clinical symptoms or positive CDI test	All	90	No CDI + antibiotic exposure (other than tetracycline) within 90 d
Dial et al, 2008 [16]	Case-control, retro	Hospital (inpatients and outpatients)	Community-acquired	Canada	1996–2004	ICD-9 codes	All	45	No CDI regardless of antibiotic exposure within 45 d
Doern-berg et al, 2012 [11]	Cohort, retro	Hospital (inpatients only)	Not available	United States	2005–2010	Diarrhea plus positive CDI toxin (ELISA)	Doxycycline	30	No CDI + ceftriaxone exposure within 30 d
Kuntz et al, 2011 [15]	Case-control, retro	Hospital (inpatients and outpatients)	Both community- and hospital-acquired	United States	2004–2007	ICD-9 codes	All	180	No CDI regardless of antibiotic exposure within 180 d
Tartof et al, 2015 [17]	Cohort, retro	Hospital (inpatients only)	Both community- and hospital-acquired	United States	2011–2012	Diarrhea plus positive CDI toxin (polymerase chain reaction)	Doxycycline	90	No CDI regardless of antibiotic exposure within 90 d

Abbreviations: CDI, *Clostridium difficile* infection; ELISA, enzyme-linked immunosorbent assay; ICD-9, *International Classification of Diseases, Ninth Revision*; retro, retrospective.

Study Setting

We performed an analysis for the setting of CDI treatment. Of the 6 studies, 3 included patients treated as inpatients only and 3 had mixed populations (inpatients and outpatients). Tetracyclines were associated with decreased risk of CDI in the studies with inpatients only (OR, 0.55; 95% CI, 0.40–0.75; $P < .001$) but not in the studies that included both inpatients and outpatients (OR, 0.92; 95% CI, 0.58–1.46; $P = .72$; Figure 6).

Difference Between Subgroups

There was no significant difference between the subgroups based on study design ($P_{\text{interaction}} = 0.71$) or CDI definition ($P_{\text{interaction}} = 0.24$). However, the subgroup analysis based on study setting revealed that the 2 subgroups were significantly different ($P_{\text{interaction}} = 0.07$). This analysis explains the significant heterogeneity seen in the overall analysis.

DISCUSSION

In this metaanalysis, we found that the use of tetracyclines was associated with a 38% lower risk of CDI compared with use of other antibiotics. Subgroup analysis of studies that evaluated the risk of doxycycline alone also revealed a protective effect. The association was consistent in cohort studies and studies with inpatients only. However, no increased or decreased risk for CDI was seen in case-control studies. The lack of a protective effect seen in this subgroup may be due to a smaller sample size. Subgroup analysis of studies with mixed populations (inpatients and outpatients) also demonstrated no increased or decreased risk of CDI; however, a protective effect was seen in the inpatient population. It is possible that the protective effect of tetracyclines might be lower in low-risk outpatients, and a higher effect size might be seen in higher-risk inpatients. Historically, antibiotic use has been considered the most important risk factor for development of CDI, with different classes of antibiotics posing various degrees of risk. Clindamycin, third-generation cephalosporins, penicillin, and fluoroquinolones are associated with a high risk of CDI [22]; tetracyclines, however, have been infrequently associated with CDI [23].

The mechanism by which tetracycline use confers this decreased risk is unclear. The decreased risk of CDI in patients who receive tetracyclines may be due to their in vitro activity against *C. difficile* [24]. In one study, 84% of *C. difficile* isolates had a minimum inhibitory concentration of ≤ 0.25 mg/L to tetracycline [24]. A European study with data from 14 countries reported increased susceptibility of *C. difficile* isolates to tetracyclines, with resistance rates of only 9.2% [25]. Studies have shown a high concentration of doxycycline in the ileum and colon in humans 4 to 6 hours after oral ingestion. High concentrations of the drug in the bowel could explain its high efficacy against *C. difficile* [26]. It is also possible that tetracyclines are less disruptive to the gut microbiota. One study that assessed the soil microbial response to application of tetracycline

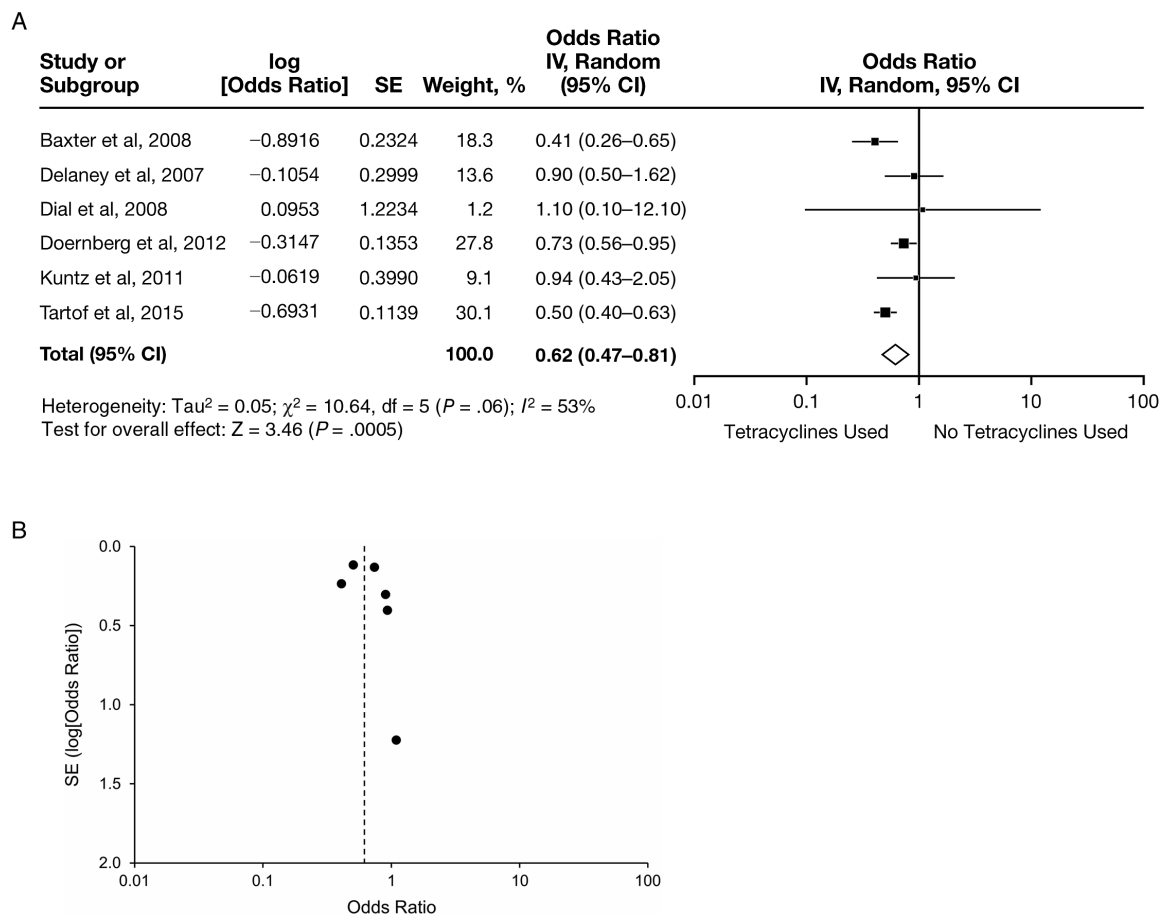


Figure 2. Analysis of all included studies. *A*, Forest plot demonstrating decreased odds of *Clostridium difficile* infection with tetracycline use by the random-effects model. *B*, Funnel plot demonstrating no publication bias. Abbreviations: CI, confidence interval; df, degrees of freedom; IV, inverse variance; SE, standard error.

concluded that tetracyclines had a shorter lasting effect on the soil microbiome than other antibiotics [27]. A recent study in mice demonstrated that the duration of disrupted gut colonization resistance was shorter for the glycylcycline antibiotic tigecycline (a minocycline derivative) than for other antibiotics [28]. Tetracyclines as protein-synthesis inhibitors may interfere with *C. difficile* toxin production and potentially decrease the risk of active infection.

Successful treatment of refractory cases of CDI with tigecycline has also been reported [29–31]. Eravacycline is an investigational novel fluorocycline with structural similarity to that of tigecycline. It has broad-spectrum activity against gram-positive organisms, gram-negative organisms, and anaerobes and is more potent than tigecycline and other tetracyclines [32]. With structural similarities to other tetracyclines, eravacycline may possibly be associated with a

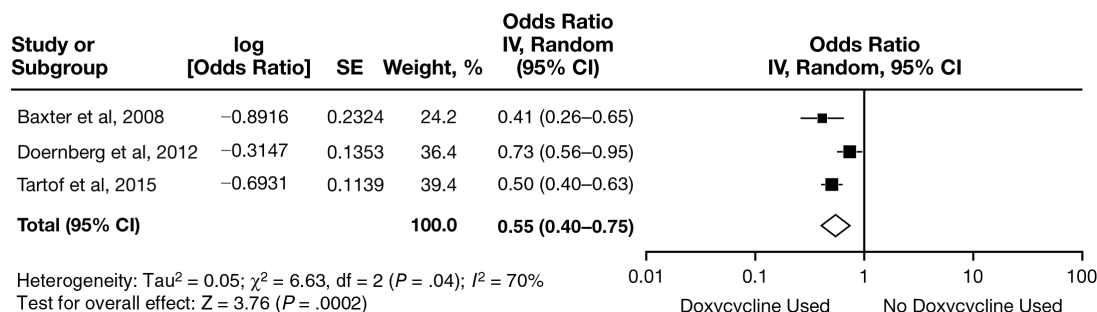


Figure 3. Forest plot of studies evaluating risk of *Clostridium difficile* infection (CDI) with doxycycline only demonstrating decreased odds of CDI with doxycycline use. IV indicates inverse variance. Abbreviations: CI, confidence interval; df, degrees of freedom; IV, inverse variance; SE, standard error.

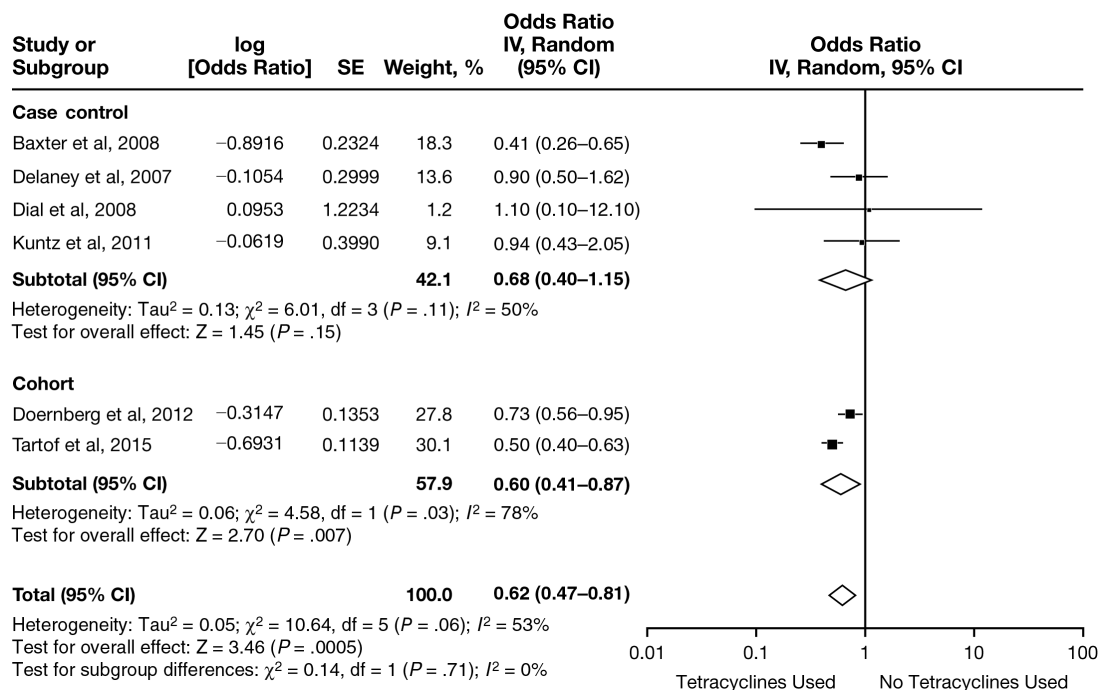


Figure 4. Forest plots by type of study demonstrating no difference in risk of *Clostridium difficile* infection (CDI) with tetracyclines in case-control studies and decreased risk of CDI with tetracyclines in cohort studies, by the random-effects model. IV indicates inverse variance. Abbreviations: CI, confidence interval; df, degrees of freedom; IV, inverse variance; SE, standard error.

decreased risk of CDI and could be a potential alternative for patients with increased risk of CDI. However, its protective role for CDI has not been studied.

Our metaanalysis focused on the role of tetracyclines in primary CDI. However, some studies have evaluated the association of CDI recurrence with tetracycline use. One study

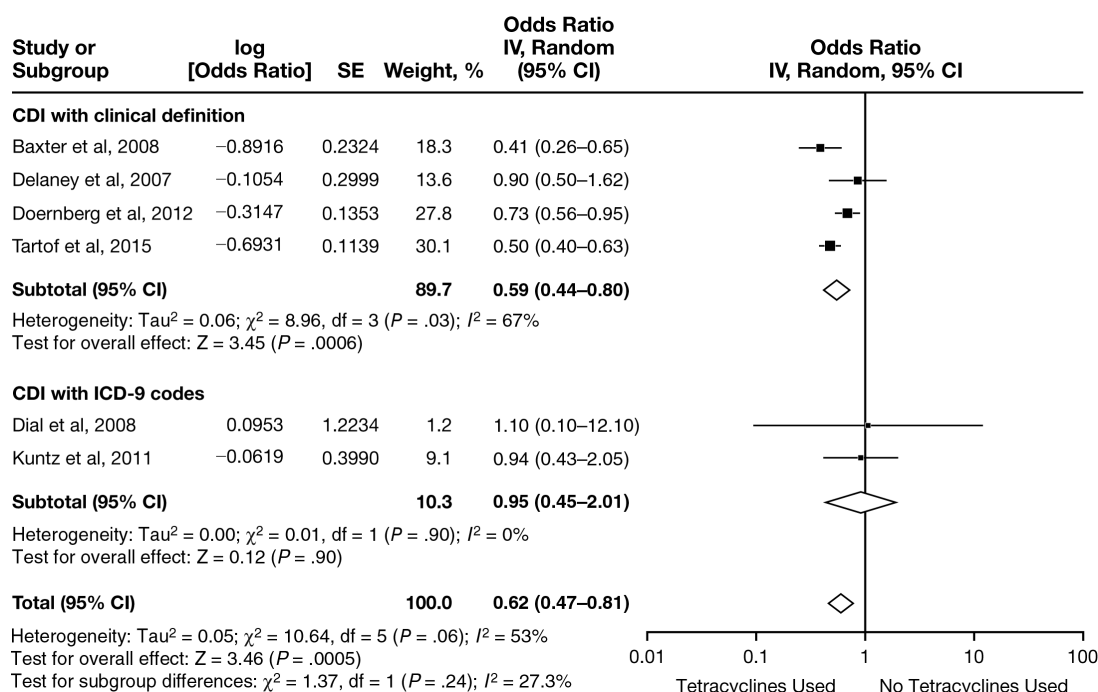


Figure 5. Forest plots by mode of *Clostridium difficile* infection (CDI) diagnosis demonstrating decreased risk of CDI with tetracycline use in studies that diagnosed CDI clinically but not in studies that used *International Classification of Diseases, Ninth Revision* codes, by the random-effects model. IV indicates inverse variance. Abbreviations: CDI, *Clostridium difficile* infection; CI, confidence interval; df, degrees of freedom; ICD-9, *International Classification of Diseases, Ninth Revision*; IV, inverse variance; SE, standard error.

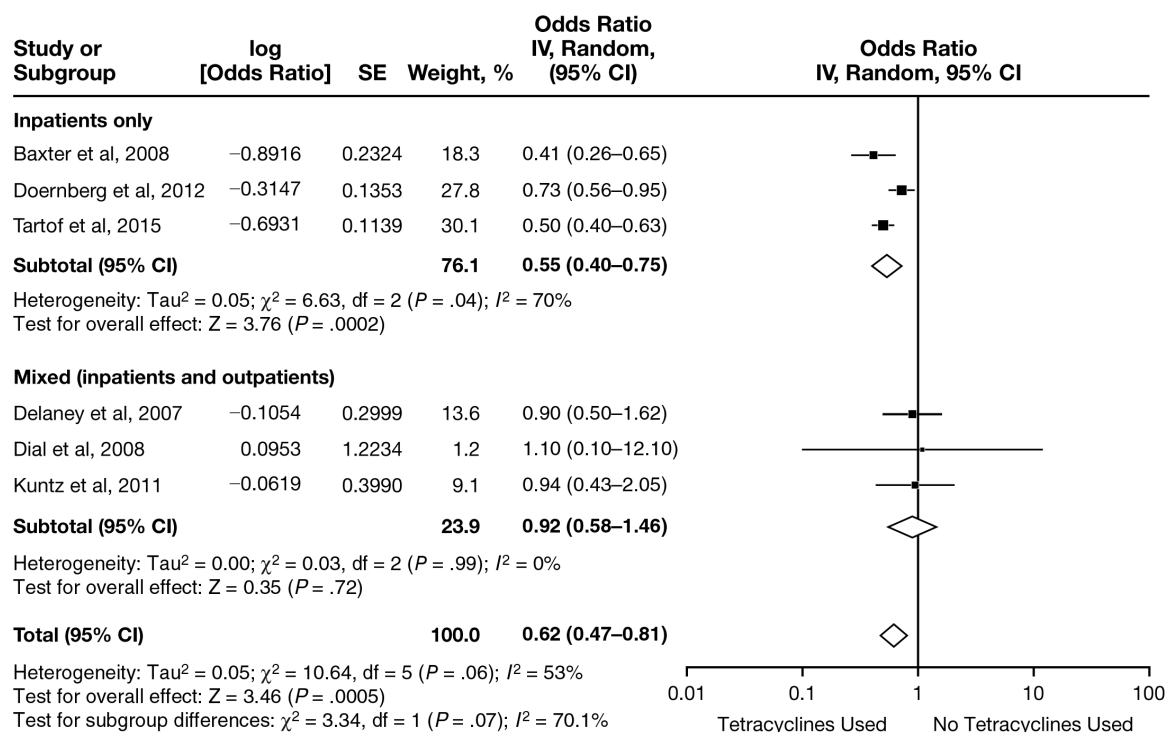


Figure 6. Forest plot by study setting demonstrating decreased risk of *Clostridium difficile* infection with tetracyclines in inpatient setting only but not in the studies that included both inpatients and outpatients, by the random-effects model. IV indicates inverse variance. Abbreviations: CI, confidence interval; df, degrees of freedom; IV, inverse variance; SE, standard error.

showed that concomitant use of tetracyclines up to 8 weeks after the initial CDI episode was associated with a protective effect for recurrent CDI (OR, 0.35; 95% CI, 0.14–0.90) [33]. Further studies assessing the risk of recurrent CDI with use of tetracyclines are needed and might be helpful in choosing antibiotics for systemic infections in patients with recurrent CDI.

Most studies included in our metaanalysis had adjusted for important confounders that are implicated in the risk of CDI (Supplementary Table S2), such as gastric acid suppression medications, age, sex, and recent hospitalization [12]. Other potential confounders including race, Charlson comorbidity index, and other comorbid conditions were handled differently in various studies. One study reported a slight increased risk of CDI with a Charlson comorbidity index of 3 or higher (OR, 1.3; 95% CI, 1.2–1.5) [17], whereas another study reported an increased risk with concomitant inflammatory bowel disease (OR, 4.1; 95% CI, 2.6–6.6) and renal failure (OR, 1.7; 95% CI, 1.3–2.2) [16].

Subgroup analysis of the studies that specifically evaluated the risk of CDI with doxycycline revealed that it was associated with a decreased risk of CDI. Doxycycline is the most commonly used tetracycline in clinical practice and has various indications [34]. In addition, doxycycline has atypical coverage against *Rickettsia*, *Borrelia*, *Chlamydia*, *Mycoplasma*, *Vibrio*, *Leptospira*, and *Entamoeba*. It has many advantages over other tetracyclines, including less-frequent dosing, improved absorption,

greater tissue distribution, longer half-life, and less photosensitivity [35–37]. Given the advantages of doxycycline, using it as a first-line therapy in place of other agents whenever possible (eg, in place of fluoroquinolones or macrolides in treatment of common infections such as community-acquired pneumonia or bronchitis) might reduce the risk of CDI. However, for urinary tract infections, abdominal infections, hospital-acquired pneumonias, and dental prophylaxis, doxycycline cannot be used as an alternative. Therefore, additional studies are necessary regarding the role of other agents that might be associated with a decreased risk of CDI in infections for which doxycycline cannot be used (eg, the use of nitrofurantoin or fosfomycin vs fluoroquinolones in urinary tract infection).

All the studies included in this metaanalysis had different time windows for antibiotic exposure before the index CDI diagnosis date, ranging from 30 to 180 days. Because the risk of CDI associated with antibiotics might decrease with increased time from the exposure, the studies with a shorter period for antibiotic exposure assessment before the index date might be associated with an increased risk of CDI. However, the study with the shortest antibiotic exposure period (30 days) before CDI showed a protective effect of doxycycline for the risk of CDI [11]. We performed a subgroup analysis based on the mode of CDI diagnosis—clinical CDI diagnosis vs administrative codes—assuming that a clinical diagnosis might be more representative of true CDI cases [38]. The risk of CDI after

treatment with tetracyclines was decreased in the studies that diagnosed CDI clinically, but no decreased or increased risk of CDI was seen among the 2 studies that used ICD-9 codes to diagnose CDI. Although ICD-9 codes have reasonable sensitivity and specificity, some case misclassification may have occurred compared with cases diagnosed clinically with laboratory confirmation [39].

The strengths of our study include a comprehensive literature search with well-defined inclusion criteria and multiple subgroup analyses, with no subgroup analysis showing an increased risk of CDI with the use of tetracyclines. Limitations of our metaanalysis include that the individual studies varied in several ways including study design, patient population (age, sex, inpatient vs outpatient status), method of CDI diagnosis, time period for prior antibiotic exposure, and dose and duration of tetracycline use. These variations led to substantial heterogeneity. Although all the studies controlled for some potential confounders, each individual study controlled for different variables. Hence, we were unable to perform analyses in which all confounding factors could be accounted or controlled for, including duration of exposure and dose of tetracyclines, indication for tetracycline use, adherence to infection-control practices, use of administrative diagnosis codes, and the possibility that a positive stool assay represented colonization rather than active infection. None of the studies included outpatients only; hence, we could not perform analyses to separate inpatients vs outpatients only. We also could not perform analyses to separate patients with hospital-acquired vs community-acquired CDI. All studies assessed the risk of primary CDI and excluded patients with recurrent CDI.

In conclusion, this study suggests that tetracycline use (particularly doxycycline) may be associated with a decreased risk of future development of CDI compared with other antibiotics. However, given the discrepancies seen on subgroup analyses, further comparative studies should be performed in settings in which tetracyclines are still effective therapeutic agents. These prospective studies should focus on the role of tetracyclines in reducing the risk of primary and recurrent CDI and comparing inpatients vs outpatients and community-acquired vs hospital-acquired CDI. Until that time, clinicians may wish to consider doxycycline over alternative antimicrobials where appropriate.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Note

Potential conflicts of interest. All authors: No reported conflicts of interest. All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

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